1.71 Variable Acceleration

Worked Solutions

1 a)

$v = \frac{\mathrm{d}s}{\mathrm{d}t} = 12t^2 - 5$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = 24t$

1 b)

$v = \frac{\mathrm{d}s}{\mathrm{d}t} = 5t^4 + 4t$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = 20t^3 + 4$

1 c)

$v = \frac{\mathrm{d}s}{\mathrm{d}t} = 2t - 1$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = 2$

1 d)

$v = \frac{\mathrm{d}s}{\mathrm{d}t} = 2 + \frac{1}{t^2}$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{2}{t^3}$

2 a)

$s = \int (8t^3 - 6t)\,\mathrm{d}t = 2t^4 - 3t^2 + c$

When $t = 0, s = 0 \Rightarrow c = 0$

$s = 2t^4 - 3t^2$

2 b)

$s = \int (6t^2 + 2t^3)\,\mathrm{d}t = 2t^3 + \frac{1}{2}t^4 + c$

When $t = 0, s = 0 \Rightarrow c = 0$

$s = 2t^3 + \frac{1}{2}t^4$

2 c)

$s = \int (\frac{1}{10}t^3 - \frac{1}{2}t^2)\,\mathrm{d}t = \frac{1}{40}t^4 - \frac{1}{6}t^3 + c$

When $t = 0, s = 0 \Rightarrow c = 0$

$s = \frac{1}{40}t^4 - \frac{1}{6}t^3$

2 d)

$s = \int (t - t^{-2})\,\mathrm{d}t = \frac{1}{2}t^2 + \frac{1}{t} + c$

When $t = 0, s = 0 \Rightarrow c = 0$

$s = \frac{1}{2}t^2 + \frac{1}{t}$

3 a)

$v = 3t^5 - 2t^2 + c$ When $t = 1, v = 2 \Rightarrow c = 1$

$v = 3t^5 - 2t^2 + 1$

3 b)

$v = 3t^3 + t^5 + c$ When $t = 2, v = 40 \Rightarrow c = -16$

$v = 3t^3 + t^5 - 16$

3 c)

$a = -0.09t^2 + 2.7t \Rightarrow v = -0.03t^3 + 1.35t^2 + c$

When $t = 1, v = 3 \Rightarrow c = 1.68$

$v = -0.03t^3 + 1.35t^2 + 1.68$

3 d)

$a = t^{\frac{1}{2}} - t^{-3} \Rightarrow v = \frac{2}{3}t^{\frac{3}{2}} + \frac{1}{2}t^{-2} + c$

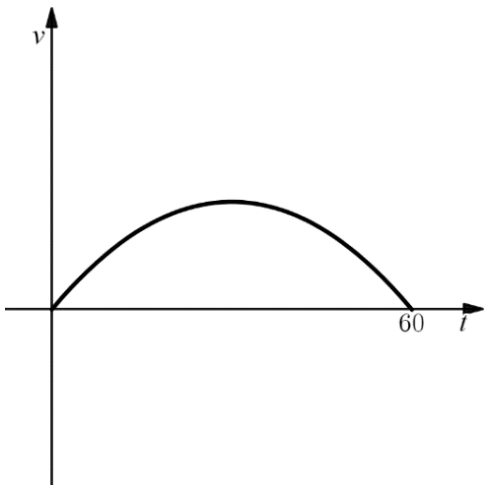
When $t = 1, v = 2 \Rightarrow c = \frac{5}{6}$ so $v = \frac{2}{3}t^{\frac{3}{2}} + \frac{1}{2}t^{-2} + \frac{5}{6}$

4 a)

Stat when $v = 0$

$0 = -\frac{1}{50}t(t - 60)$ so at $t = 0$ and $t = 60$

4 b)



4 c)

$\int_0^{60} -\frac{1}{50}t^2 + \frac{6}{5}t \, \mathrm{d}t = \left[-\frac{1}{150}t^3 + \frac{3}{5}t^2\right]_0^{60}$

$= (720) - (0) = 720$ m

4 d)

$a = -\frac{1}{25}t + \frac{6}{5}$

4 e)

The model suggest that the car will continue with only a negative velocity after 60 seconds which would mean the car reversing from that point on.

5 a)

$v = \int (15t^{\frac{1}{2}} + 2)\,\mathrm{d}t = 10t^{\frac{3}{2}} + 2t + c$

When $v = 1, t = 0$ so $1 = 10 \times 0^{\frac{3}{2}} + 2 \times 0 + c \Rightarrow c = 1$

$v = 10t^{\frac{3}{2}} + 2t + 1$

5 b)

$s = \int (10t^{\frac{3}{2}} + 2t + 1)\,\mathrm{d}t = 4t^{\frac{5}{2}} + t^2 + t + c$

When $s = 6, t = 1$ so $6 = 4 \times 1^{\frac{5}{2}} + 1^2 + 1 + c \Rightarrow c = 0$

$s = 4t^{\frac{5}{2}} + t^2 + t$

6 a)

$v = \int -30t^2 + 25t^3\,\mathrm{d}t = -10t^3 + \frac{25}{4}t^4 + c$

At $t = 0, v = 5$ so $5 = 0 + c \Rightarrow c = 5$

$v = -10t^3 + \frac{25}{4}t^4 + 5$

6 b)

$s = \int -10t^3 + \frac{25}{4}t^4 + 5\,\mathrm{d}t = -\frac{5}{2}t^4 + \frac{5}{4}t^5 + 5t + c$

At $t = 0, s = 0$ so $s = -\frac{5}{2}t^4 + \frac{5}{4}t^5 + 5t$

$s|_{t=2} = -40 + 40 + 10 = 10$ m

7 a)

At $(4, 8)$, $8 = -\frac{3}{8} \times 16 + 4k + 2 \Rightarrow k = 3$ as req.

7 b)

$s = 8$ m

7 c)

$2 = -\frac{3}{8}t^2 + 3t + 2 \Rightarrow t = 0, t = 8$ s

8 seconds

7 d)

$v = -\frac{3}{4}t + 3$

7 e)

$v = -\frac{3}{4} \times 10 + 3 = -4.5$ ms^{-1}

So speed $= 4.5$ ms^{-1}

8 a)

$s = \int v\,\mathrm{d}t = 4t^2 - t^3 + c$

When $t = 0, s = 0 \Rightarrow c = 0$ so $s = 4t^2 - t^3$

8 b)

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = 8 - 6t$

8 c)

$F = 0.4(8 - 6t) = 3.2 - 2.4t$

9 a)

$v = kt - 3 + t^{-2}$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = k - 2t^{-3} \Rightarrow 1.75 = k - 0.25 \Rightarrow k = 2$

9 b)

$\int_{\frac{1}{2}}^{10} 2t - 3 + t^{-2}\,\mathrm{d}t = \left[t^2 - 3t - \frac{1}{t}\right]_{\frac{1}{2}}^{10}$

$= \left(100 - 30 - \frac{1}{10}\right) - \left(\frac{1}{4} - \frac{3}{2} - 2\right) = 73.15$ m

10 a)

$s = -\frac{1}{k}t^2 + \frac{240}{k}t \Rightarrow v = -\frac{2}{k}t + \frac{240}{k}$

$4 = -\frac{2}{k} \times 20 + \frac{240}{k} \Rightarrow 4k = -40 + 240$

$\Rightarrow k = 50$ as req

10 b)

$v = -\frac{1}{25}t + \frac{24}{5} \Rightarrow v|_{t=60} = -\frac{12}{5} + \frac{24}{5} = 2.4$ ms^{-1}

10 c)

$v|_{t=120} = -\frac{24}{5} + \frac{24}{5} = 0$ ms^{-1}

10 d)

Ellis has reached a speed of 0 ms^{-1} so he will have a negative velocity which means he has changed direction.

11)

$s = \int v\,\mathrm{d}t = -\frac{1}{2}t^2 + 10t + c$

At $t = 0, s = 0$ so $s = -\frac{1}{2}t^2 + 10t$

$0 = -\frac{1}{2}t^2 + 10t \Rightarrow t = 0, t = 20$

So 20 seconds

12 a)

$v = \int a\,\mathrm{d}t = 4t + \frac{3}{2}t^2 - 2t^3 + c$

When $t = 0, v = 0 \Rightarrow c = 0$

$v = 4t + \frac{3}{2}t^2 - 2t^3$

12 b)

$s = \int v \, \mathrm{d}t = 2t^2 + \frac{1}{2}t^3 - \frac{1}{2}t^4 + c$

When $t = 1, s = 6 \Rightarrow c = 4$

$s = 2t^2 + \frac{1}{2}t^3 - \frac{1}{2}t^4 + 4$

12 c)

$0 = 2t^2 + \frac{1}{2}t^3 - \frac{1}{2}t^4 + 4 \Rightarrow t = 2.80$ s (3 SF)

13 a)

$s|_{t=10} = 520$ m

13 b)

$v = 3t^2 - 12t + 12$

At rest when $v = 0, t = 2$

13 c)

$v = 3t^2 - 12t + 12 = 3(t-2)^2 \geq 0$ for all t

So it never reverses direction.

13 d)

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = 6t - 12$ which is not constant.

14)

$v = \int a \, \mathrm{d}t = -0.9t^2 + 18t + c$

At $t = 0, v = 0 \Rightarrow c = 0$

$v = -0.9t^2 + 18t$

$s = \int v \, \mathrm{d}t = -0.3t^3 + 9t^2 + c$

At $t = 0, s = 0 \Rightarrow c = 0$

$s = -0.3t^3 + 9t^2$

$600 = -0.3t^3 + 9t^2$ so $t = 10, 10 \pm 10\sqrt{3}$

It takes 10 seconds to travel 600 m

15 a)

$\int_0^{20} 2t - 0.001t^3 \, \mathrm{d}t = \left[t^2 - \frac{1}{4000}t^4\right]_0^{20}$

$= (400 - 40) - (0) = 360$ m

15 b)

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = 2 - 0.003t^2$

a has max of 2 when $t = 0$,

When $t = 25, a = 0.125$ so $a \geq 0$ for $0 \leq t \leq 25$ so always accelerating.

16 a)

$v = \frac{\mathrm{d}s}{\mathrm{d}t} = \frac{8}{15}t - \frac{1}{225}t^2$

16 b)

$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{8}{15} - \frac{2}{225}t$ so max when $\frac{8}{15} - \frac{2}{225}t = 0$

At $t = 60$ seconds.

So $v = 32 - 16 = 16$ ms^{-1}

17 a)

$12.5 = 25a + 125b$ and $25 = 100a + 1000b$

Gives $a = 0.75, b = -0.05$

17 b)

$s = \int_0^{10}(0.75t^2 - 0.05t^3)\,\mathrm{d}t = \left[0.25t^3 - \frac{1}{80}t^4\right]_0^{10}$

$= (250 - 125) - (0) = 125$ m

$s = \frac{1}{2}(0 + 25) \times 10 = 125$ m which is the same.

18 a)

$v = \frac{3.5}{7}t = \frac{1}{2}t$

18 b)

$\frac{1}{2} \times 7 \times 3.5 = 12.25$ metres

18 c)

$0 = -\frac{1}{k}(-7)^2 + 3.5 \Rightarrow k = 14$

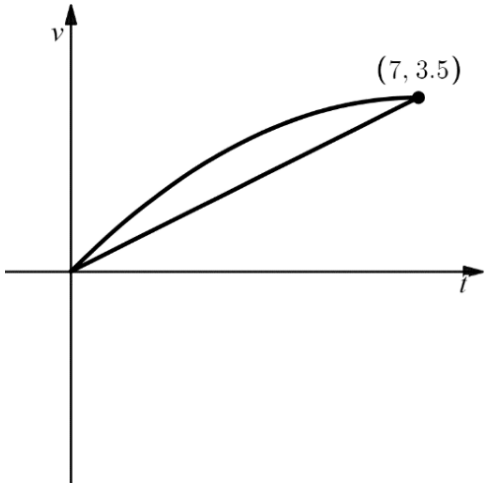
18 d)

$v = -\frac{1}{14}(t - 7)^2 + 3.5 = -\frac{1}{14}t^2 + t$

$\int_0^7 -\frac{1}{14}t^2 + t\,\mathrm{d}t = \left[-\frac{1}{42}t^3 + \frac{1}{2}t^2\right]_0^7$

$= \left(-\frac{49}{6} + \frac{49}{2}\right) - (0) = \frac{49}{3} = 16.3$ metres

18 e)



The area under the curve between $t = 0$ and $t = 7$ is larger for the second (quadratic) model.

19 a)

$s = \int_0^5 \frac{2}{5}t^3 - 4t^2 + 10t \, \mathrm{d}t = \left[\frac{1}{10}t^4 - \frac{4}{3}t^3 + 5t^2\right]_0^5$

$= \left(\frac{125}{2} - \frac{500}{3} + 125\right) - (0) = \frac{125}{6} = 20.8$ m (3 SF)

19 b)

$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{6}{5}t^2 - 8t + 10$ stat pt when $\frac{\mathrm{d}v}{\mathrm{d}t} = 0$ so $t = \frac{5}{3}, 5$ s

$\therefore t = \frac{5}{3}$ s which gives $v = \frac{200}{27} = 7.41$ ms^{-1} (3 SF)

19 c)

$a = \frac{6}{5}t^2 - 8t + 10$ has a minimum at $t = \frac{10}{3}$ giving $a = -\frac{10}{3}$ so minimum acceleration is -3.33 ms^{-2}

This is a deceleration of 3.33 ms^{-2}

19 d)

As $t \to \infty, v \to \infty$ which is not possible. That is, the cyclist will not be able to increase in velocity (with increasing acceleration) permanently.

20 a)

$v = kt^{-\frac{1}{2}} + \frac{1}{80}t \Rightarrow a = -\frac{1}{2}kt^{-\frac{3}{2}} + \frac{1}{80}$

$0.0065 = -\frac{1}{250}k + \frac{1}{80} \Rightarrow k = \frac{3}{2}$ as req.

20 b)

$\int_{60}^{120} \frac{3}{2}t^{-\frac{1}{2}} + \frac{1}{80}t \, \mathrm{d}t = \left[3t^{\frac{1}{2}} + \frac{1}{160}t^2\right]_{60}^{120}$

$= (3\sqrt{120} + 90) - \left(3\sqrt{60} + \frac{45}{2}\right)$

$= 77.1$ m (3 SF)

20 c)

For smaller values of t, the velocity is very large which is unrealistic.

21)

From initial conditions: $r = 7.8$

$3 = 16p + 4q + 7.8 \Rightarrow 16p + 4q = -4.8$

Min when $\frac{\mathrm{d}v}{\mathrm{d}t} = 2pt + q = 0$ so $8p + q = 0$

$p = 0.3, q = -2.4$

22)

$v = \int a \, \mathrm{d}t$ so $v = \frac{1}{2}kt^2 - 4t + c$

$v|_{t=0} = 2 \; v|_{t=2}$

$v|_{t=0} = c$ and $v|_{t=2} = 2k - 8 + c$

$c = 4k - 16 + 2c$

$16 - 4k = c$

$v = \frac{1}{2}kt^2 - 4t + 16 - 4k$

$= \frac{1}{2}kt^2 - 4(t + k) + 16$ as req.

Min velocity when $\frac{\mathrm{d}v}{\mathrm{d}t} = 0$ so $0 = 2k - 4$

$k = 2$ which gives $v = t^2 - 4t + 8$

23)

$v = 0 \Rightarrow 0 = \frac{3}{2}t^2 - \frac{9}{2}t + 3 \Rightarrow t = 1, t = 2$

$s = \int v \, \mathrm{d}t = \frac{1}{2}t^3 - \frac{9}{4}t^2 + 3t + c$

When $t = 0, s = 0 \Rightarrow c = 0$

$s = \frac{1}{2}t^3 - \frac{9}{4}t^2 + 3t$

$s|_{t=1} = 1.25$ and $s|_{t=2} = 1$

$8.5 - 0.25 - 0.25 = 8$ metres

24 a)

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = -2t + k \Rightarrow 6 = 0 + k$ so $k = 6$

24 b)

Max v when $\frac{\mathrm{d}v}{\mathrm{d}t} = 0$ so $0 = -2t + 6 \Rightarrow t = 3$

24 c)

$s = \int v \, \mathrm{d}t$ so $s = \int_0^3 -t^2 + 6t \, \mathrm{d}t = \left[-\frac{1}{3}t^3 + 3t^2\right]_0^3$

$= (-9 + 27) - (0) = 18$ metres

24 d)

$\int_0^9 -t^2 + 6t \, \mathrm{d}t = \left[-\frac{1}{3}t^3 + 3t^2\right]_0^9$

$= (-243 + 243) - (0) = 0$ metres

24 e)

$v = 0 \Rightarrow 0 = -t^2 + 6t \Rightarrow t = 6$ as $t > 0$

24 f)

$\int_6^9 -t^2 + 6t \, \mathrm{d}t = \left[-\frac{1}{3}t^3 + 3t^2\right]_6^9$

$= (-243 + 243) - (-72 + 108) = -36$

Total distance is $18 + 18 + 36 = 72$ metres

25 a)

$v = 0$

25 b)

The cyclist has increasing velocity for the first 6 seconds before cycling with constant velocity of 9.72 ms^{-1} for the next 14 seconds.

25 c)

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = 1.62t - 0.27t^2$

Max when $\frac{\mathrm{d}a}{\mathrm{d}t} = 0$ so $0 = 1.62 - 0.54t$

$\Rightarrow t = 3$ seconds

$a|_{t=3} = 2.43$ ms^{-2}

26 a)

$800 = \frac{3}{200}t^2 - \frac{1}{40000}t^3 \Rightarrow t = -200, t = 400$

So 400 seconds.

26 b)

$0 = -3.2t + 2080 \Rightarrow t = 650$ so 650 seconds in total.

26 c)

$v = \frac{\mathrm{d}s}{\mathrm{d}t} = \frac{3}{100}t - \frac{3}{40000}t^2$

Max v when $\frac{\mathrm{d}v}{\mathrm{d}t} = 0$ so $0 = \frac{3}{100} - \frac{3}{20000}t \Rightarrow t = 200$

So $v = \frac{3}{100} \times 200 - \frac{3}{40000} \times 200^2 = 3$ ms^{-1}

26 d)

On the way back: $v = 3.2$ ms^{-1}

So max is 3.2 ms^{-1}

26 e)

It is unlikely that Filip ran with a fixed velocity instantaneously from leaving the shop to arriving exactly back at his house.

27 a)

$s = 6 \times 5.56 - 15 + 15e^{-0.4\times 5.56} = 19.9826331$

≈ 20 m as req

27 b)

$v = 6 - 6e^{-0.4t}$

27 c)

$v|_{t=0} = 6 - 6 = 0$ ms^{-1}

27 d)

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = 2.4e^{-0.4t}$ for $0 \leq t \leq 6$

So max at $t = 0 \Rightarrow a = 2.4$ ms^{-2}

28 a)

$s|_{t=0} = 0.8 + 8 = 8.8$ m above ground level

28 b)

Change in direction when $\frac{\mathrm{d}s}{\mathrm{d}t} = 0$

$0.08e^{0.1t} - 1.6e^{0.2t} = 0 \Rightarrow 0.08e^{0.1t} = 1.6e^{-0.2t}$

$\Rightarrow 20 = e^{0.3t} \Rightarrow \ln(20) = 0.3t \Rightarrow t = 9.98577\ldots$

So at $t = 9.99$ seconds (3 SF)

28 c)

At $t = 9.96, s = 3.25730114\ldots$ so 3.26 metres (3 SF)

28 d)

$v = 0.08e^{0.1t} - 1.6e^{-0.2t}$

28 e)

$v|_{t=0} = 0.08 - 1.6 = -1.52$ ms^{-1}

$v|_{t=40} = 4.37$ ms^{-1}

Stat pt when $\frac{\mathrm{d}v}{\mathrm{d}t} = 0$

$0.008e^{0.1t} + 0.32e^{-0.2t} = 0 \Rightarrow 0.008e^{0.1t} = 0.32e^{-0.2t}$

$\Rightarrow 40 = e^{0.3t} \Rightarrow \ln(40) = 0.3t \Rightarrow t = 12.29626485\ldots$

So at $t = 12.3$ seconds (3 SF)

$v|_{t=12.3} = 0.137$ ms^{-1}

So max speed is 4.37 ms^{-1}

28 f)

$s = 0 + \frac{1}{2} \times 9.8 \times 2^2 = 19.6$ metres

29)

$s = -\frac{1}{10}t^3 + \frac{2}{5}t^2 \Rightarrow v = \frac{\mathrm{d}s}{\mathrm{d}t} = -\frac{3}{10}t^2 + \frac{4}{5}t$

$\Rightarrow a = \frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{3}{5}t + \frac{4}{5}$

$F = 0.2 \times \left(-\frac{3}{5}t + \frac{4}{5}\right) = -\frac{3}{25}t + \frac{4}{25}$